GlucoTwin

A Digital Twin that warns of glucose excursions 2 hours ahead by fusing EHR and wearable / CGM data (Type 2 Diabetes)

Happiest Health · Reimagining and Reforming Healthcare in India Summit 2026 · Digital Twin Challenge

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Why diabetes needs a twin, not a snapshot

Millions

of people live with diabetes in India

Minutes

of doctor time every few months, per patient

5 min

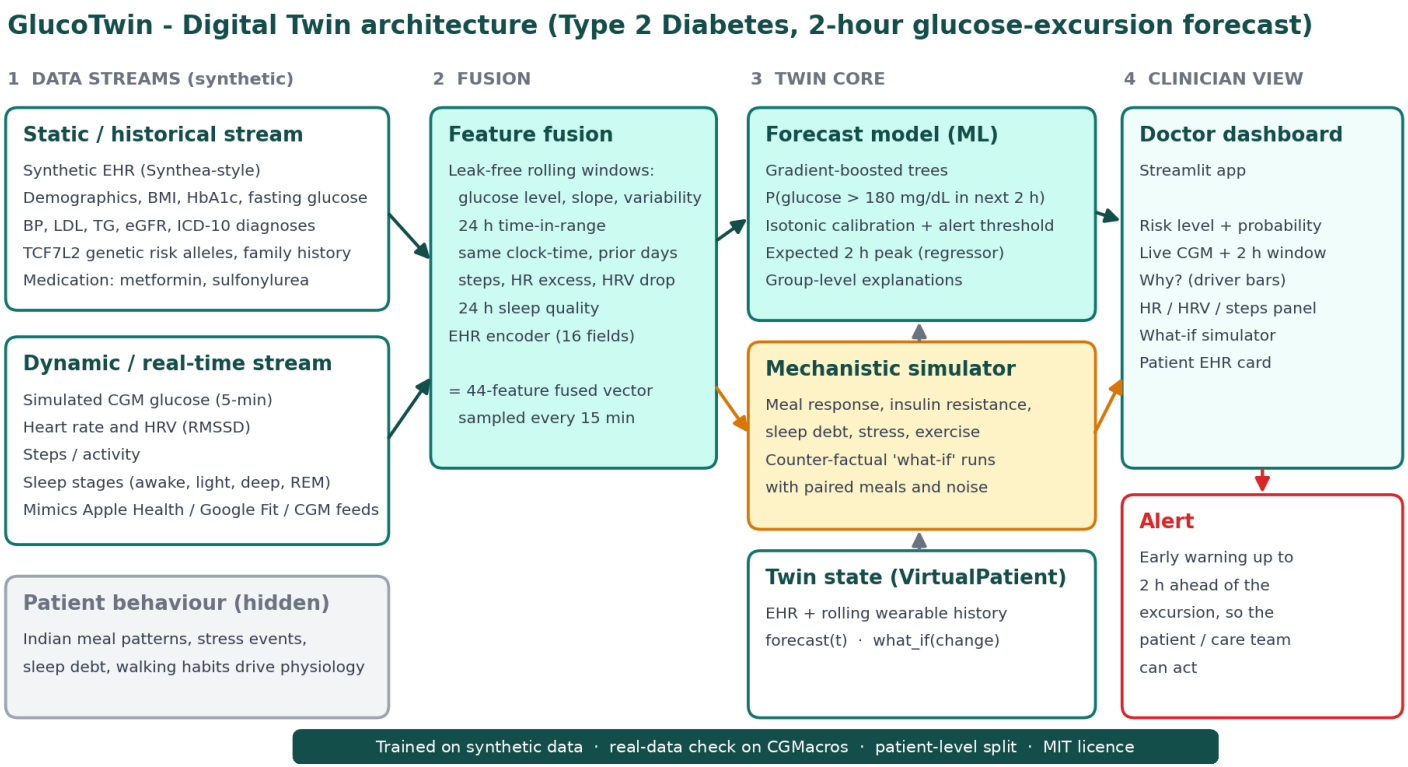
how often glucose, heart rate and sleep data change on a wearable

Today's care reacts after the spike. A Digital Twin keeps a living virtual replica of each patient, updated by wearables and joined with the health record, so the care team can act before an adverse event.

Use case: forecast, explain, test a fix

Condition	Type 2 Diabetes
Adverse event	Glucose above 180 mg/dL at any time in the next 2 hours (currently at or below 160)
User	Diabetologist or care team looking at a virtual patient console
Output	Probability + expected peak + risk level + 'why' + what-if simulation

Architecture



Two data streams, simulated for training

Static / historical (EHR)

- Age, sex, BMI, HbA1c, fasting glucose
- BP, LDL, triglycerides, eGFR
- ICD-10 diagnoses, medications
- TCF7L2 genetic risk alleles, family history

Dynamic / real-time (wearables)

- CGM glucose every 5 minutes
- Heart rate and HRV (RMSSD)
- Steps and activity
- Sleep stages (awake, light, deep, REM)

150 virtual patients × 14 days × 5-minute resolution. Indian meal patterns, sleep debt, stress and walking habits drive the physiology. Training uses no real patient data. The real open dataset CGMacros (de-identified, PhysioNet) is used only for validation and is not redistributed.

The twin is a hybrid: data-driven + mechanistic

1

Forecast (ML)

Gradient-boosted trees on 44 fused, leak-free features. Calibrated probability, expected 2 h peak, alert threshold tuned for about 80% sensitivity.

2

What-if (simulator)

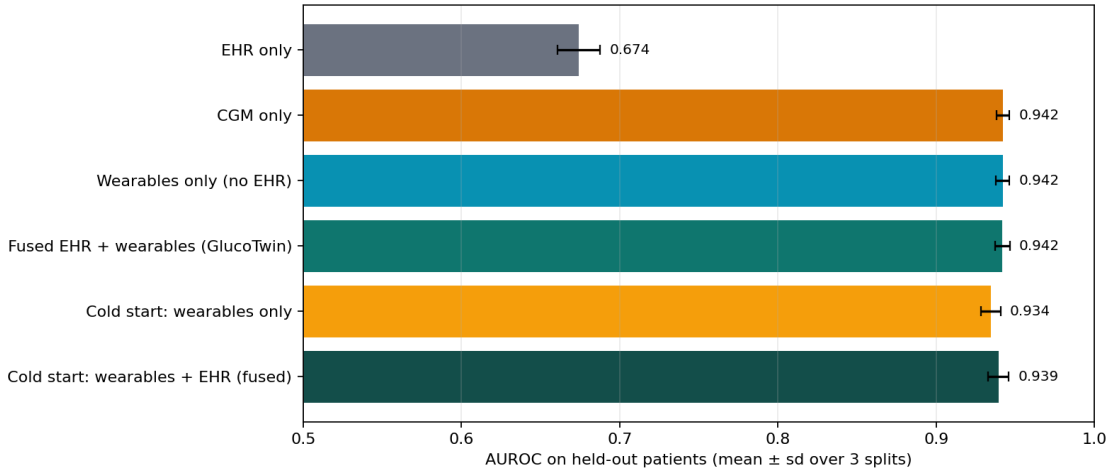
Meal response, insulin resistance, sleep debt, stress and exercise. Re-runs the same meals and noise with one change. Insulin resistance is estimated from the patient's first 5 days of CGM (tested in simulation only).

3

Explain (drivers)

Each forecast shows which input group pushes risk up or down: glucose trend, meal-time habit, sleep, clinical profile, stress, movement.

Results on held-out synthetic patients



0.942

AUROC, fused model (3 patient-level splits)

98%

pre-warned (primary split), at ~6 alerts per patient-day

EHR alone: 0.67. Logistic regression on glucose + trend: 0.61. Prevalence about 17%.

Labels come from our simulator, so this measures pipeline learning, not clinical accuracy.

Honest reading of the results

1

Glucose history leads

Recent CGM and meal-time habit carry most of the signal (importance 0.17 and 0.12 AUROC). EHR alone cannot time a spike (AUROC 0.67).

2

Fusion helps only at cold start, not with history

With CGM history: AUPRC 0.771 (CGM), 0.769 (wearables), 0.770 (fused), so no gain. Cold start: 0.739 to 0.762, within one sd (3 splits). On real data, adding EHR fields gave no gain (AUROC 0.827 vs 0.817).

3

HRV, sleep and steps add little to the forecast

Importance only about 0.001 to 0.006 AUROC in simulation, even though they drive the simulated physiology. They mainly add explanatory context for the clinician.

Real-data check: CGMacros, 45 adults

Setting	AUROC (95% CI)	AUPRC
A. Synthetic-trained, tested on real	0.826 (0.795–0.853)	0.422
B. Real-trained, participant-grouped CV	0.827 (0.795–0.852)	0.454
B + 7 shared EHR fields	0.817 (0.783–0.842)	0.446
A, T2D only (n=14)	0.835 (0.811–0.856)	0.554
B, T2D only (n=14)	0.820 (0.799–0.835)	0.577

15 healthy, 16 prediabetes, 14 T2D; about 10 days each; Dexcom G6 Pro; 25,306 forecast times; event prevalence 12.1%. CGM and time-of-day features only. CI resamples whole participants.

0.94 → ~0.83

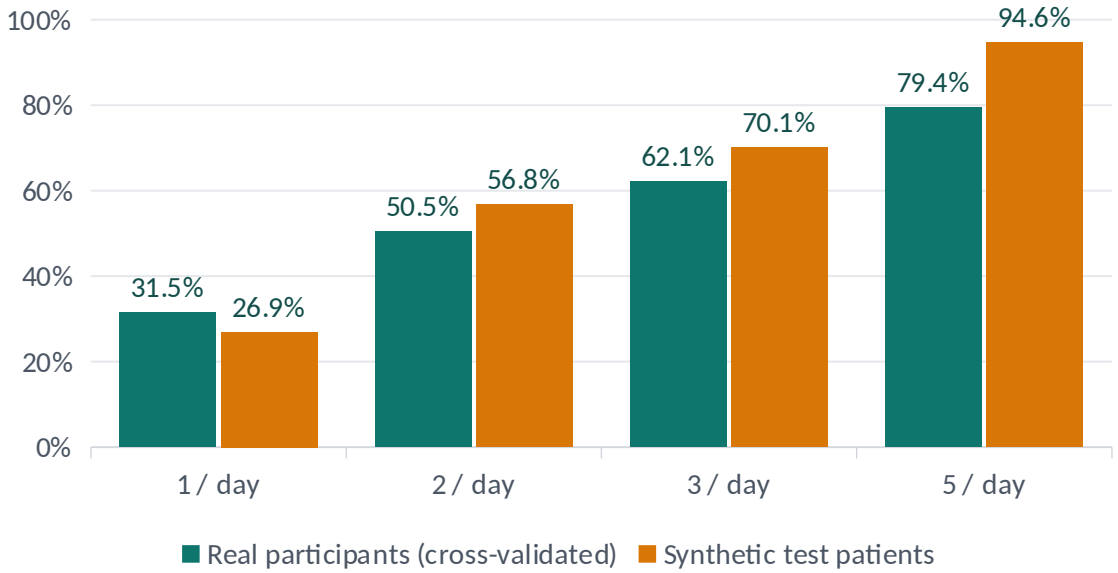
AUROC, synthetic patients vs real participants (sim-to-real gap)

No gain

from EHR fusion on real data (45 participants, 7 shared EHR fields)

One US site, mostly mild T2D, no HRV/sleep/steps in this dataset: not clinical evidence. Libre readings gave a higher AUROC (about 0.89 to 0.90, prevalence 5.4%, values possibly interpolated); Dexcom was chosen in advance as primary.

Alert burden: episodes warned vs alerts per day



~50%

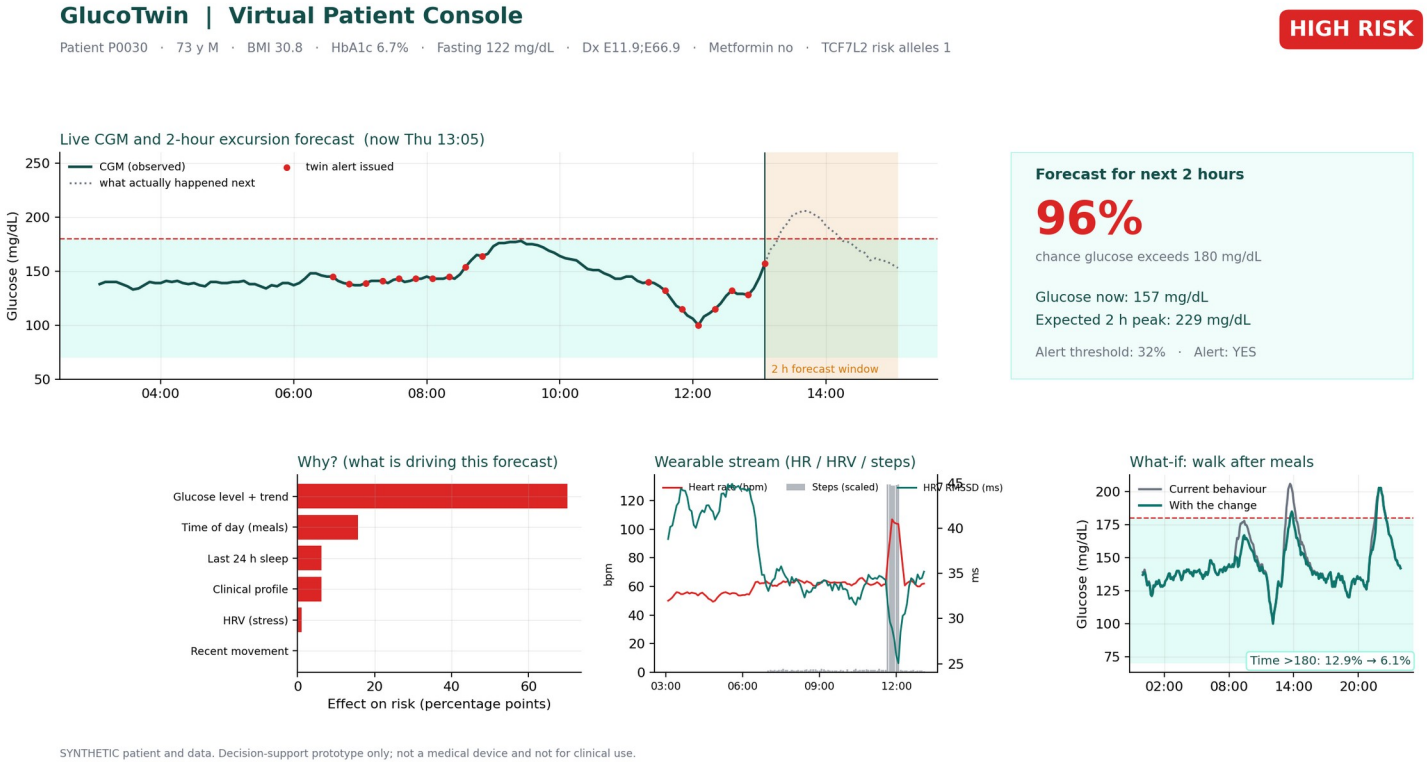
of episodes warned at 2 alerts per patient-day (real participants)

97.9%

warned at the original ~5.8 alerts per day (synthetic only)

Proposed two-tier design: a continuous risk bar on the dashboard, push alerts only at a low alert budget. Thresholds were picked on the same predictions they are reported on, so figures are illustrative; the operating point should be set with clinicians.

Conceptual UI: the doctor's console



What-if: walk 15 minutes after meals



12.9% → 6.1%

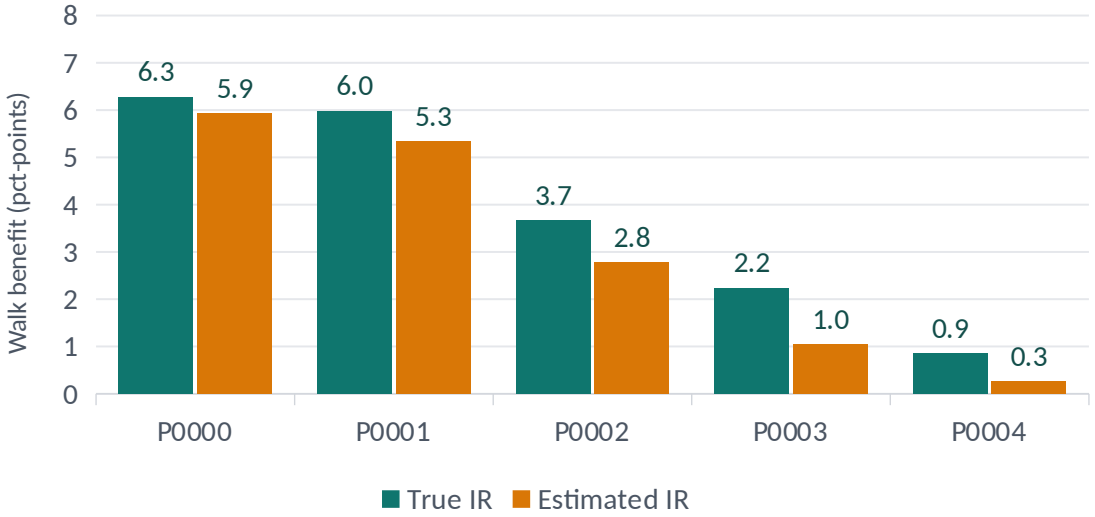
time above 180 mg/dL (demo patient P0030, true insulin resistance)

231 → 205

mean daily peak, mg/dL

Same meals, same noise, one change. Effect size reflects simulator assumptions, not clinical evidence. With insulin resistance estimated from CGM the benefit is smaller (next slide).

Twin calibration: estimating insulin resistance



r = 0.88

estimated vs true IR, 150 synthetic patients (first 5 days of CGM)

0.094

IR error (MAE) vs 0.164 for the cohort-mean baseline; true IR sd 0.196

In all 5 patients checked, the walk benefit was underestimated with estimated IR (small example). P0003's estimate hit the grid edge (0.30).

Optimistic: the same simulator generates and fits the data. Walk habit, resting HR, meal timings and noise seed are still true values. Not tested on real patients.

Limitations and responsible use

- Trained on synthetic data. Real check: 45 adults, one US site, mostly mild T2D; not clinical evidence
- Sim-to-real gap: AUROC 0.94 on synthetic, about 0.83 on real. EHR fusion gain not seen on real data
- Results depend on the CGM sensor (Dexcom 0.83 primary; Libre about 0.89 to 0.90, possibly interpolated)
- Twin calibration estimates only insulin resistance and was tested in simulation only (optimistic); other parameters are true values
- Fixed 180 mg/dL threshold; alert-fatigue risk; operating points not chosen on separate data
- What-if results are simulations, not prescriptions. Research prototype, not a medical device, no regulatory clearance

Next steps

- Validate on larger real datasets (e.g. ShanghaiT2DM) and anonymised clinical records, with ethics approval
- Estimate more twin parameters from each patient's own data (walk habit, meal timing), and test on real patients
- Clinician-tuned alert policy: two-tier design, with the alert budget set by diabetologists
- Add uncertainty intervals and re-test sensor dependence (Dexcom vs Libre)

Thank you · github.com/subhrank09/DigitalTwiner_BITMESRA